

BASIC idea:  $\dot{\underline{x}} = A \underline{x}$ ,  $\underline{x} \in \mathbb{R}^n$

If we know any  $n$ -sol's, from linearly indep. ICs, we know all sol's.

= More formally, given  $\{\underline{\psi}_k(t)\}_{k=1}^n$  sol's with  $\{\underline{\psi}_k(t_0)\}_{k=1}^n$  linearly indep., then from any ICs  $\underline{x}_0 = \underline{x}(t_0)$ ,

$$\Rightarrow \underline{x}(t) = \sum_{k=1}^n \alpha_k \underline{\psi}_k(t), \alpha_k \in \mathbb{R}$$

$\{\underline{\psi}_k(t)\}_{k=1}^n$  are called "fund'l sol's"

Define fund. matrix:  $\underline{\Psi}(t) = [\underline{\psi}_1(t) \dots \underline{\psi}_n(t)]$  with  $\underline{\Psi}(t_0)$  nonsingular.

$$\underline{x}(t_0) = \underline{x}_0 = \underline{\Psi}(t_0) \underline{\alpha} \quad \underline{\alpha} = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}$$

$$\Rightarrow \underline{\alpha} = \underline{\Psi}^{-1}(t_0) \underline{x}_0$$

$$\underline{x}(t) = \underline{\Psi}(t) \underline{\alpha} \quad \forall t \geq t_0$$

=

$$\frac{d}{dt} \underline{\Psi}(t) = A(t) \underline{\Psi}(t)$$

$\underline{\Psi}(t_0)$  nonsingular

Hence

$$\underline{x}(t) = \underline{\Phi}(t, t_0) \underline{x}_0$$

$\underline{\Phi}(t, t_0)$  "state transition matrix"

$$\underline{x}(t) = \underline{\Phi}(t, t_0) \underline{x}_0$$

Note:

$$\underline{\alpha} = \underline{\Psi}(t_0)^{-1} \underline{x}_0 = \underline{\Psi}(t)^{-1} \underline{x}(t) \quad \text{for all } t \geq t_0$$

and we showed in fact  $\underline{\Psi}(t)$  is invertible "  
= what if  $\underline{u}(t) \neq 0$ ? Is  $\underline{\alpha}$  still "conserved"

Note

$$\frac{d}{dt} [\underline{\alpha}] = \underline{0} = \frac{d}{dt} [\underline{\Psi}^{-1}(t) \underline{x}(t)] \quad \text{when } \underline{u}(t) = \underline{0}$$

when  $\underline{u}(t) \neq 0$

$$\frac{d}{dt} [\underline{\Psi}^{-1} \underline{x}] = \left[ \frac{d}{dt} \underline{\Psi}^{-1} \right] \underline{x} + \underline{\Psi}^{-1} \dot{\underline{x}}$$

$$= \left[ \frac{d}{dt} \underline{\Psi}^{-1} \right] \underline{x} + \underline{\Psi}^{-1} [A \underline{x} + B \underline{u}]$$

what is  $\frac{d}{dt} \underline{\Psi}^{-1}$

$$\frac{d}{dt} [\psi \psi^{-1} = \mathbb{I}]$$

$$\left( \frac{d}{dt} \psi \right) \psi^{-1} + \psi \left( \frac{d}{dt} \psi^{-1} \right) = 0$$

$$\Rightarrow \frac{d}{dt} \psi^{-1} = -\psi^{-1} [A \psi] \psi^{-1}$$

$$\frac{d}{dt} \psi^{-1} = -\psi^{-1} A$$

so

$$\frac{d}{dt} [\psi^{-1} \underline{x}] = (-\psi^{-1} A) \underline{x} + \psi^{-1} [A \underline{x} + B \underline{u}]$$

$$\Rightarrow \frac{d}{dt} [\psi^{-1} \underline{x}] = \psi^{-1} B \underline{u}$$

Integrate both sides

$$\int_{t_0}^t \frac{d}{d\tau} [\psi^{-1}(\tau) \underline{x}(\tau)] d\tau = \int_{t_0}^t \psi^{-1}(\tau) B(\tau) \underline{u}(\tau) d\tau$$

$$= [\psi^{-1}(\tau) \underline{x}(\tau)]_{\tau=t_0}^t = \int_{t_0}^t (\cdot) d\tau$$

so

$$\psi^{-1}(t) \underline{x}(t) - \psi^{-1}(t_0) \underline{x}(t_0) = \int_{t_0}^t \psi^{-1}(\tau) B(\tau) \underline{u}(\tau) d\tau$$

so that

$$\underline{x}(t) = \underbrace{\psi(t) \psi^{-1}(t_0)}_{\Phi(t, t_0)} \underline{x}_0 + \int_{t_0}^t \underbrace{\psi(t) \psi^{-1}(\tau)}_{\Phi(t, \tau)} B(\tau) \underline{u}(\tau) d\tau$$

so finally

$$\underline{x}(t) = \underbrace{\Phi(t, t_0) \underline{x}_0}_{\underline{x}_{zi}(t)} + \underbrace{\int_{t_0}^t \Phi(t, \tau) B(\tau) \underline{u}(\tau) d\tau}_{\underline{x}_{zs}(t)}$$

$\underline{x}_{zi}(t)$

"zero input"  
response  
(due only to  $\underline{x}_0$ )

$\underline{x}_{zs}(t)$

"zero (initial) state"  
response  
(due only to  $\underline{u}(t)$ )

Similar decomposition for output

$$y(t) = C(t)x(t) + D(t)u(t) \\ = \underbrace{C(t)x_{zi}(t)}_{y_{zi}(t)} + \underbrace{C(t)x_{zs}(t) + D(t)u(t)}_{y_{zs}(t)}$$

$$y_{zi}(t) = C(t)\Phi(t, t_0)x_0$$

$$y_{zs}(t) = D(t)u(t) + \int_{t_0}^t C(t)\Phi(t, \tau)B(\tau)u(\tau)d\tau$$

Equivalently

$$y_{zs}(t) = \int_{t_0}^t G(t, \tau)u(\tau)d\tau$$

$$G(t, \tau) = p \times m \text{ "impulse response matrix"}$$

$$G(t, \tau) = C(t)\Phi(t, \tau)B(\tau) + D(\tau)\delta(t - \tau)$$

where  $\delta(t - \tau)$  is the  
"delta function" (unit impulse)

which has the property that  
for any  $f'$ ,  $f(t)$ ,

$$\int f(\tau)\delta(t - \tau)d\tau = f(t) \\ \text{("sifting property")}$$

$$\equiv \text{Rewrite general state sol'n as} \\ x(t) = \Psi(t) \left[ \Psi^{-1}(t_0)x_0 + \int_{t_0}^t \Psi^{-1}(\tau)B(\tau)u(\tau)d\tau \right]$$

$$\text{or } x(t) = \Psi(t)\alpha(t) \\ = \sum_{k=1}^p \alpha_k(t)\psi_k(t)$$

Note

$$\dot{\alpha}(t) = \Psi^{-1}(t)B(t)u(t)$$

$$\frac{d}{dt} x(t) = \frac{d}{dt} [\Psi(t)\alpha(t)]$$

$$= \dot{\Psi}(t)\alpha(t) + \Psi(t)\dot{\alpha}(t)$$

$$= A(t)\Psi(t)\alpha(t) + \cancel{\Psi(t)}\cancel{\Psi^{-1}(t)}B(t)u(t)$$

$$\dot{x}(t) = A(t)x(t) + B(t)u(t) \quad \checkmark$$

$\Rightarrow$  if we know  $\Psi(t)$  (or  $\Phi(t, \tau)$ )  
we know everything about system.  
What can we say about these  
generally?

$\Phi(t, \tau)$  is an  $n \times n$  matrix,  
defined by  $n^2$  functions of  $\tau, t$   
Usually will not know these f's  
exactly, and must argue using only  
generic properties of any STM.

### General properties of any STM

- 1.)  $\Phi(t, t) = \Phi(\tau, \tau) = I$  for  
any  $t, \tau$ .
- 2.) for any  $t, \tau$ ,  $\Phi(t, \tau)$  is nonsingular  
(... to be continued!)